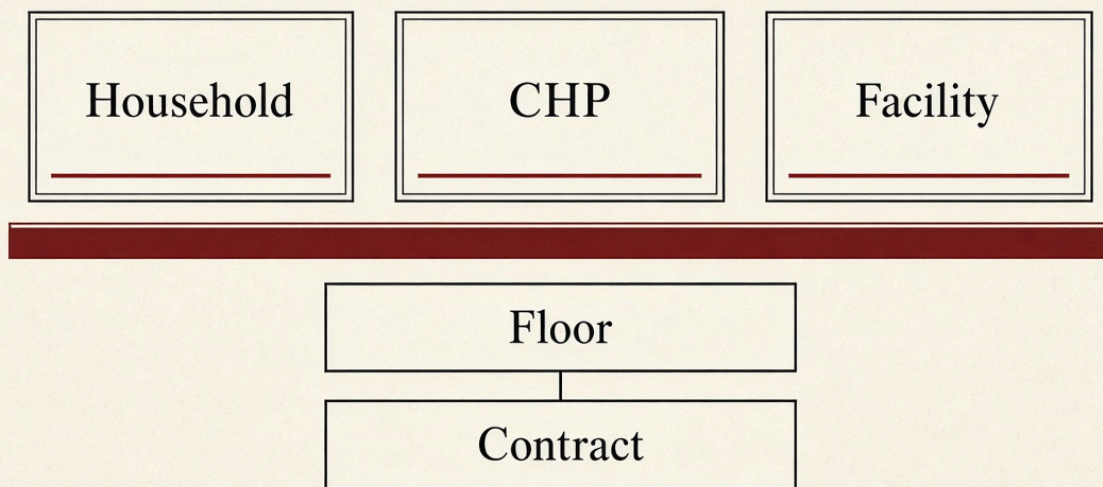


# Plan: Tunza architecture as developed

One public Next.js app. A demo floor that cannot be downgraded. A ten-section contract that is not yet on the screen.

## ► Metadata



Three surfaces. One floor. The contract sits under the verdict, not instead of it.

## Purpose

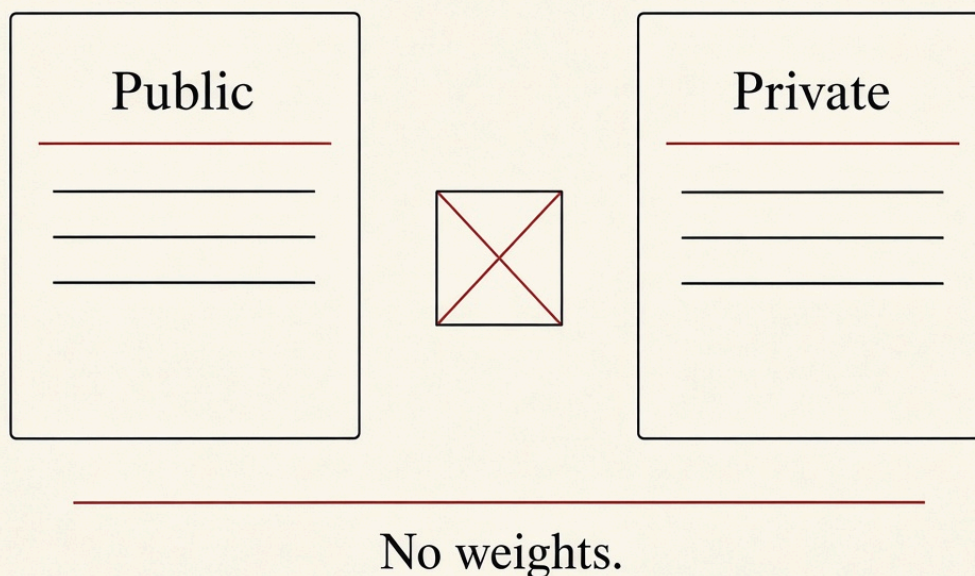
Record the Tunza architecture that actually exists on 2026-10-07, and specify the one bounded slice that finishes it: put the clinical contract on the existing care path without opening training, retrieval, the private prompt lane, or a release.

This is an engineering plan. It is not a clinical approval, not a device claim, and not a model release. The demo rules in `lib/assessment.ts` stay labeled not clinically validated.

## Problem

The pitch and the second-brain page still talk as if Tunza were either unbuilt or already a trained healthcare model. The code is neither. The public repo is a working three-surface demo. The private repo is a hosted prompt in front of Anthropic, with no floor under it. Wave 1 built the missing contract in the public tree, then stopped before the screen used it.

- There are no owned weights, no adapter, and no training set in either tree. The phrase "existing Tunza clone model" means the private prompt lane, not a clone. Source: `vault/capability-map.md` section 0.
- The public decision is `decide()` at `lib/assessment.ts:91`. The store still calls it directly at `lib/store.tsx:198`. `runClinicalPipeline` is not imported by any component.
- The README "What exists today" list includes vitals, case history, outbreak signals, verified clinicians, notifications, web push, and a PWA. Those are not in the public code. Code is truth. Source: `vault/capability-map.md` section 17.
- Host recovery wrote the rest of wave 1 into the working tree and did not commit or push. Head of the last child commit is `8228512`. Source: `vault/wave1-handoff.md`.
- The second-brain page `wiki/tunza.md` still says the product is pre-build and that the architecture names are absent from `~/code/Tunza`. That page is stale relative to this tree. Do not treat it as the current architecture.



Two products. No weights. The unsafe junction is the private lane with nothing under the model.

## Solution

Keep the public app as the trunk. The demo floor remains the only approved path. A candidate that is lower, unrecognized, or merely higher-but-unapproved is rewritten to the floor and recorded as a disagreement. The ten sections stay separate. The household still sees one verdict. Nothing in this slice trains, retrieves, encodes a guideline threshold, edits `medical-triage`, or authorizes a release.

### SURFACES

Household, CHP, and facility. Same referral object. Copy from `lib/copy.ts` only. `describeReferral` is the only place referral state becomes words.

### PRODUCT SPINE

Reducer in `lib/store.tsx`. Verdict is still a `Decision` from `decide()`. Persistence stays `localStorage` key `tunza.v2.care`. Do not store the contract there.

### FLOOR

`decide()`. Four kinds: go now, get care today, monitor at home, need one more answer. English and Kiswahili danger keywords. Abstain by asking, not by inventing certainty.

### CONTRACT

`lib/clinical/contract.ts`. Ten sections. Disposition table adds `escalate` and `abstain` internally. No new user-visible headline. `abstain` has a null headline key.

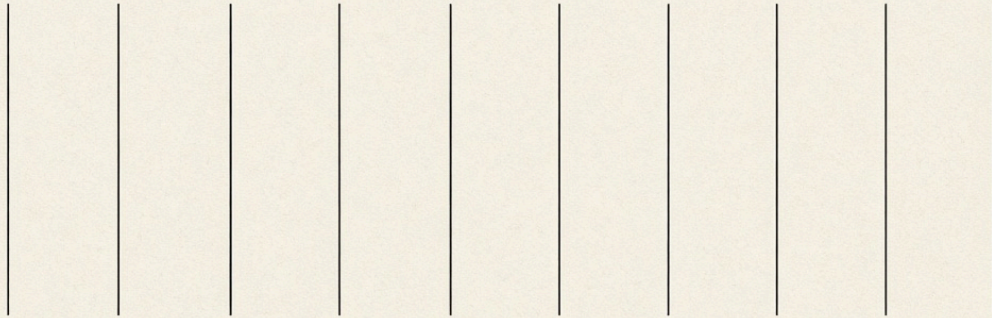
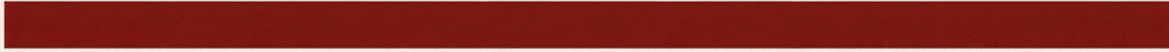
### STATE AND RECONCILE

Encounter-scoped append-only events. Same-type conflicts stay unresolved. Trends are structural, not clinical cutoffs. Reconciler fail-closed: `fail_open` is the literal `false`.

### EVAL

Family-keyed reports. Retrieval, calibration, clinical correctness, multilingual generation, and medication safety stay `not_measurable`. No `score` field gates a release. Promotion stays blocked.

# Floor wins,



Ten sections,

The floor wins. Ten sections stay ten sections.

## Relevant Files

### Existing Files

- **EXISTING** `lib/assessment.ts` — demo floor. Do not edit the rules or `decide()`.
- **EXISTING** `lib/types.ts` — `Role`, `DecisionKind`, `ReferralStage`, `Encounter`. Read-only this slice.
- **EXISTING** `lib/copy.ts` — EN/SW table. Contract section labels already exist at the English keys around line 359 and the Kiswahili keys around line 701. A missing Kiswahili string is a compile error.
- **EXISTING** `lib/referral.ts` — canonical lifecycle and the single presentation function.
- **EXISTING** `lib/store.tsx` — reducer. Line 198 sets `decision` from `decide(answers)` when the question flow stops.
- **EXISTING** `lib/clinical/contract.ts` — runtime-validated ten-section assessment. Committed in `2a1812d`. Do not rewrite it.
- **EXISTING** `lib/clinical/state.ts` — append-only encounter projection. Working tree, uncommitted.

- **EXISTING** `lib/clinical/reconcile.ts` — floor wins. Working tree, uncommitted.
- **EXISTING** `lib/clinical/pipeline.ts` — `decide()` then reconciler then contract. Not called by the UI.
- **EXISTING** `evals/runner.ts` — frozen baseline runner. Expected labels are not derived from `decide()`.
- **EXISTING** `evals/gates.ts` — fail-closed promotion. `promotion` is the literal `"blocked"`.
- **EXISTING** `components/DecisionResult.tsx` — verdict, one action, optional `children`, optional why-list.
- **EXISTING** `components/surfaces/HouseholdSurface.tsx` — renders the verdict from `decision.kind` at line 101. Wiring point.
- **EXISTING** `components/surfaces/ChpSurface.tsx` — same component, clinical context allowed underneath.
- **EXISTING** `components/CarePath.tsx` — referral words only through `describeReferral`.
- **EXISTING** `tests/assessment.test.tsx` — locked floor cases. Must stay unchanged and green.
- **EXISTING** `vault/wave1-contract.md` — scope, deferrals, file ownership, exit checks.
- **EXISTING** `vault/wave2-ledger.md` — closed doors and the evidence required to open them.

## New Files

- **NEW** `components/ContractSections.tsx` — renders the ten `RenderedSection` rows. No prose collapse. No new copy keys unless both languages are added together.
- **NEW** `tests/contract-sections.test.tsx` — EN and SW labels, empty evidence shown as empty, headline unchanged.

## Implementation Phases

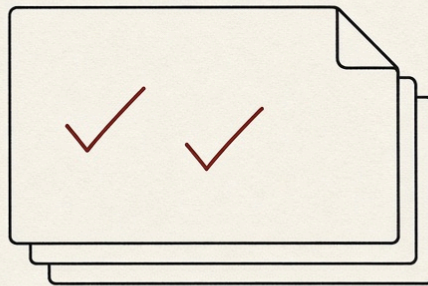
**IMPORTANT:** Execute every phase and task step by step, in order, top to bottom.

Status markers: `[]` idle · `[wip]` in progress · `[x]` complete · `[f]` failed. All start as `[]`; the Build Plan workflow updates them as it works.

## **[1] Phase 1: Re-verify the spine already in the tree**

Do not rebuild wave 1. Confirm the working tree still matches the handoff, then re-run the four gates before any UI edit. If a gate fails, fix the regression in the new wiring only after this baseline is green. Do not "repair" the floor to make a later test pass.

# Verify spine.



Verify first. The spine is already written.

### **1. Inventory the branch**

**[1]** Confirm the repo is `/Users/evanmotovich/code/Tunza` and the branch is `clinical-contract-wave1`.

**[1]** Confirm `HEAD` is `8228512` unless a later local commit on this same branch is recorded in an amendment. Do not reset, rebase, or push.

**[1]** Confirm these uncommitted paths still exist and were not rewritten:

`lib/clinical/state.ts`, `lib/clinical/reconcile.ts`, `lib/clinical/pipeline.ts`,  
`evals/gates.ts`, `evals/adversarial/`, `vault/wave1-handoff.md`, `vault/wave2-  
ledger.md`.

**[1]** Confirm `lib/clinical/contract.ts` was not edited after `2a1812d`. If it differs, stop and record the diff. Do not silently accept a rewrite.

**[1]** Confirm `decide()`, the `DecisionKind` union, and `tests/assessment.test.ts` are unchanged from `main` at `2a42af2` plus the wave-1 commits. Do not edit them in this

plan.

## 2. Re-state the invariants before touching UI

- [ ] A fixture missing any of the ten `SECTION_KEYS` fails `validateClinicalAssessment`.
- [ ] Temperatures 37.4 then 39.2 at different timestamps both remain. Encounter state has no single `temperature` field.
- [ ] Candidate disposition `self-care` against floor `go_now` reconciles to `go_now` with reason `candidate_below_floor`.
- [ ] A higher unapproved candidate is not applied. Reason is `candidate_not_approved`. The returned disposition is still the floor.
- [ ] Demo-path `model_id` and `knowledge_version` are null. Retrieved evidence is empty with `trust_tier: "none"`.
- [ ] `evaluateReleaseGate` returns `promotion: "blocked"` and `release_authorization: "none"`.

## 3. Testing Strategy

Use the existing toolchain only. No new dependencies. These commands prove the spine still matches the 2026-10-06 handoff before any surface change. If a command fails, fix the cause and re-run. Do not delete a failing assertion to go green.

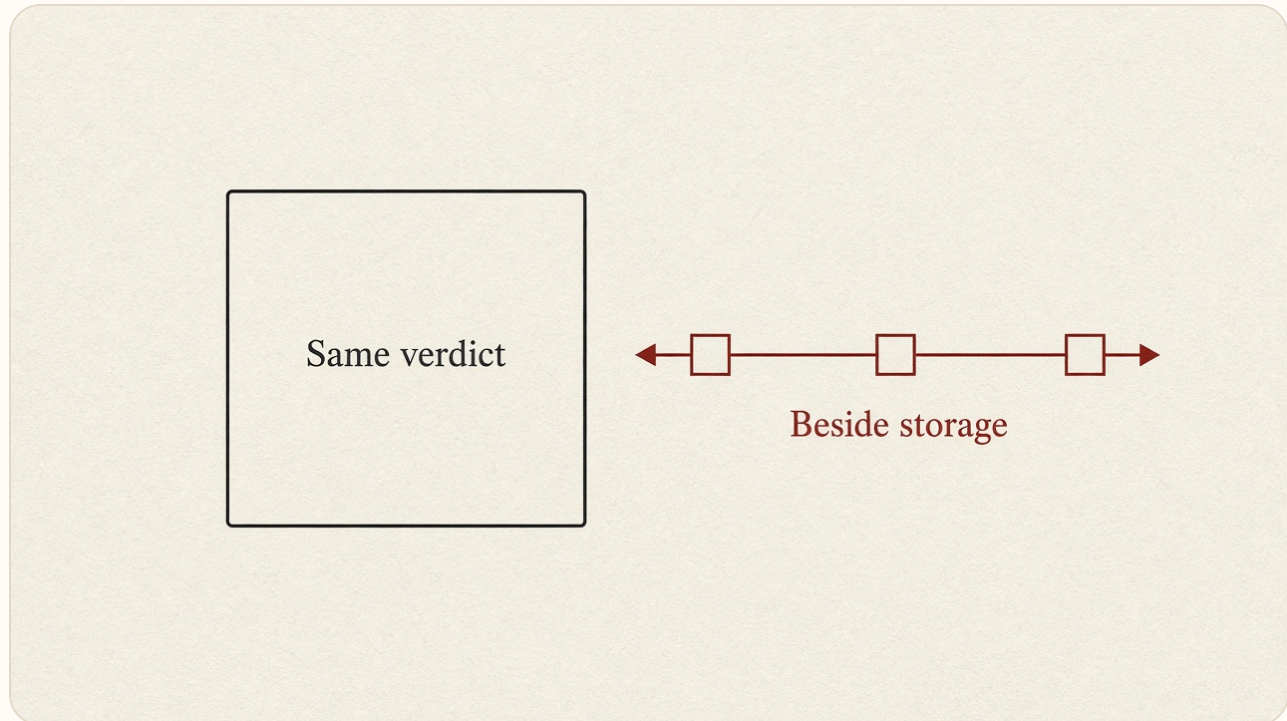
- [ ] `npx tsc --noEmit` — the contract, pipeline, and copy table still typecheck, including Kiswahili parity.
- [ ] `npm test` — prior assessment cases stay green, including Kiswahili danger keywords, child-not-drinking to `go_now`, and incomplete story to `need_one_more_answer`.
- [ ] `npm run lint` — no new lint errors in the clinical modules.
- [ ] `npm run build` — Next.js still compiles the three routes and the client surfaces.



**Do not exit this phase until every box above is checked.** If any command fails, fix the cause and re-run — loop until all pass.

## **[1] Phase 2: Call the pipeline at render time**

The gap is not a missing module. `runClinicalPipeline` exists and the store ignores it. Compute the contract from the encounter the reducer already holds. Do not change what gets persisted.



Same verdict. The pipeline runs beside the store, not inside localStorage.

## **4. Keep the reducer decision line**

**[1]** Leave `lib/store.tsx` answer handling as `const decision = upcoming ? null : decide(answers)`. The persisted `Encounter.decision` stays a `Decision`.

**[1]** Do not add `ClinicalAssessment`, `ClinicalEvent[]`, or disagreement records to `CareState`. The localStorage write would expand stored clinical detail before the privacy projections exist.

**[1]** Do not pass a model candidate from the UI. The demo path calls the pipeline with `candidate` omitted, so `model_id` stays null.

## **5. Add a pure view helper**

**[1]** Add a function, in `lib/clinical/pipeline.ts` or a sibling that only imports it, that accepts the current `Encounter` and locale and returns `PipelineResult`.

[ ] Pass `generatedAt: encounter.updatedAt`. That field is already an ISO timestamp. Do not call `new Date()` inside render; the pipeline throws if the timestamp is not explicit ISO.

[ ] If `encounter.decision` is null, do not run the pipeline. The question flow is still open.

[ ] After the call, assert `result.assessment.next_action.disposition === encounter.decision.kind`. If it differs, render the existing `decide()` verdict and do not show the contract. A mismatch is a defect, not a new headline.

[ ] Demo path must not emit `escalate` or `abstain`. `pipelineDecider` already throws on those. The view helper must do the same and fall back to the stored `Decision`.

## 6. Testing Strategy

Vitest, no browser, no network. Prove the view helper is a projection of the floor, not a second decision maker.

[ ] `npm test -- tests/clinical-pipeline.test.ts tests/assessment.test.ts` — existing floor cases and pipeline cases still pass, and `tests/assessment.test.ts` is unmodified.

[ ] `npm test -- tests/clinical-reconcile.test.ts` — `self-care` against `go_now` still rewrites up, and a telemetry throw does not change the disposition.



**Do not exit this phase until every box above is checked.** If any command fails, fix the cause and re-run — loop until all pass.

### [ ] Phase 3: Show the ten sections without a new verdict

The household screen already has one job: say what to do next. The contract is support. It must not become a second essay, a chat transcript, or a collapsed paragraph.

## One verdict

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One verdict. Ten rows underneath. Not one generated paragraph.

### 7. Build ContractSections from the existing component set

**[ ]** Create `components/ContractSections.tsx`. Props: `sections` from `renderAssessment`, plus `locale`. No new visual language. Paper card, line border, label type, same radius as `DecisionResult`.

**[ ]** Render one row per `SECTION_KEYS` entry, in that order. Use `section.label` from `t()`. Do not hard-code English.

**[ ]** Render `section.body` as structured fields. Arrays stay lists. Empty arrays stay visible as empty, including hypotheses. Do not join the ten bodies into one string.

**[ ]** Retrieved evidence renders `retrieval_status` and `trust_tier` only. Do not invent a citation. `items` is an empty tuple in this wave; the type forbids items.

**[ ]** Contradictions render claim, source, importance, and `resolution_status: "unresolved"`. Do not add a resolve button.

**[ ]** Provenance renders `rules_version`, null `model_id`, null `knowledge_version`, and `generated_at`. Do not label this "validated".

### 8. Mount it where the verdict already lives

**[ ]** In `HouseholdSurface`, keep `headline={decisionHeadline(decision.kind, locale)}`. Do not switch the headline to `rendered.headline` unless a test proves they are the same string for all four `DecisionKind` values in both locales.

**[ ]** Put `ContractSections` in the `children` slot of `DecisionResult`, inside a `details` element, closed by default. The open verdict, the one action, and the warning signs stay above it. Design rule: one dominant answer.

**[ ]** In `ChpSurface`, mount the same component expanded under the same `DecisionResult`. CHP may see the sections without an extra click. The headline is still the household headline, not a different clinical sentence.

**[ ]** Do not mount it on `FacilitySurface`. That surface speaks referral state through `CarePath` and `describeReferral`. A second clinical paragraph there would split the truth.

**[ ]** Do not add a headline for `abstain`. If the helper throws, the existing four headlines remain the only ones on screen.

**[ ]** Reuse copy keys `contractObservedFacts` through `contractProvenance`. If a new string is required, add it to both `en` and `sw` in the same edit. `sw` is `Record<CopyKey, string>`.

## 9. Testing Strategy

Testing Library plus the existing contrast and copy tests. Prove language parity and that the verdict node is unchanged.

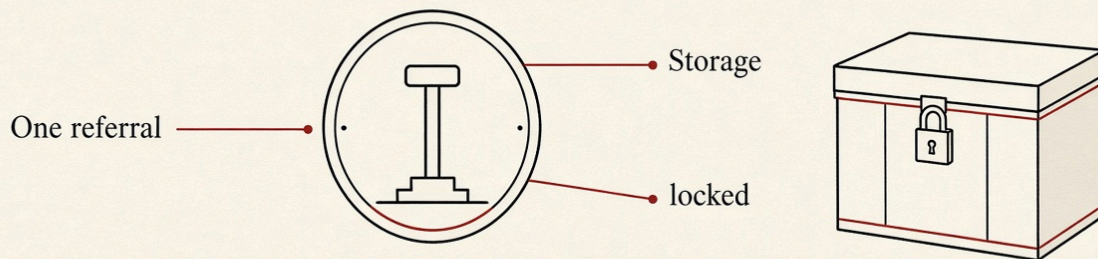
**[ ]** `npm test -- tests/contract-sections.test.tsx tests/components.test.tsx tests/copy.test.ts` — ten labels render in EN and SW, empty evidence is shown as empty, and existing component copy does not drift.

**[ ]** `npm test -- tests/contrast.test.ts` — no new raw colors. Components keep using tokens. Oxblood and emergency red stay at least 1.5× luminance apart.

 **Do not exit this phase until every box above is checked.** If any command fails, fix the cause and re-run — loop until all pass.

## **[ ] Phase 4: Leave the product boundaries where they are**

Wiring the contract must not quietly complete the README. Several neighboring systems look unfinished. They are closed on purpose.



One referral. Three views. Storage stays the current device key.

## 10. Do not cross these boundaries

- [1]** Do not edit `describeReferral` or invent referral copy in a surface. Internal stage names never appear on screen.
- [1]** Do not attach Google Places results to `referral.facilityId`. Nearby lookup is real. Referral facilities stay the labeled demo list until capability truth exists.
- [1]** Do not change `/api/facilities`, `/api/transcribe`, or `/api/access`. Precise coordinates stay on device. Worker codes stay server-side. Voice still degrades server, then on-device, then type.
- [1]** Do not add vitals, image interpretation, case history, outbreak signals, web push, or a PWA in this slice. Image attachment remains a boolean the rules do not read.
- [1]** Do not edit `/Users/evanmotovich/code/medical-triage`. The adapter ticket at `vault/tickets/medical-triage-adapter.md` stays not started.
- [1]** Do not edit anything under fusion-harness, including `adws/`. Do not add a training endpoint, a dataset writer, a weight path, a retriever, or an IMCI/ETAT threshold.
- [1]** Do not vendor guideline text. Do not put PHI, secrets, or `.env` contents in fixtures, docs, or logs.

## 11. Testing Strategy

Diff and test boundaries. The proof is absence as much as presence.

**[ ]** `git diff --stat -- lib/referral.ts lib/assessment.ts lib/types.ts app/api tests/assessment.test.ts` — those paths show no edits from this slice.

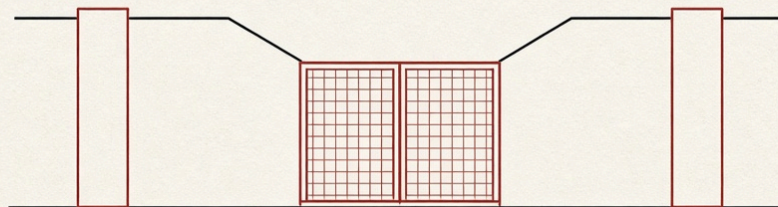
**[ ]** `git -C /Users/evanmotovich/code/medical-triage status --short` — no new edits from this work. Pre-existing dirt stays untouched.

**[ ]** `npm test -- tests/referral.test.ts tests/api.test.ts tests/places.test.ts` — referral words, access gate, and the location-blind facility proxy still pass.

 **Do not exit this phase until every box above is checked.** If any command fails, fix the cause and re-run — loop until all pass.

### **[ ] Phase 5: Exit fail-closed**

Green engineering gates are not a release. This phase records what was wired, re-runs the full gate set, and leaves promotion blocked.



Gates green

Release blocked

Gates can be green. Release stays blocked.

## **12. Write the session note, not a release note**

**[ ]** Add `vault/sessions/2026-10-07-architecture-wired.md`. State what was verified, which files changed, and that clinical effectiveness is not established.

**[ ]** Update `vault/INDEX.md` current-state paragraph. Do not claim the README capabilities that section 17 of the capability map already marked missing.

**[ ]** Do not edit `vault/wave2-ledger.md` except to append a dated line if a door's evidence actually appeared. None of those doors open because this plan's tests passed.

**[ ]** Local commit is allowed by the wave-1 contract. Push, merge, and rebase are not. Do not commit unless the four gates in the next task are green. Do not push under any result.

### 13. Testing Strategy

Full local gates plus the release function. The release function must still refuse.

**[ ]** `npx tsc --noEmit` — view helper and new component typecheck against the existing contract.

**[ ]** `npm test` — full vitest run, including unchanged assessment cases.

**[ ]** `npm run lint` — clean.

**[ ]** `npm run build` — production build still generates the static pages.

**[ ]** `npm test -- tests/eval-gates.test.ts` — promotion remains blocked, required families that are `not_measurable` still block, and no field named `score` or `accuracy` is a ship gate.



**Do not exit this phase until every box above is checked.** If any command fails, fix the cause and re-run — loop until all pass.

## Validation Commands

Execute these commands from `/Users/evanmotovich/code/Tunza` to validate the entire plan is complete:

**[ ]** `npx tsc --noEmit` — types, copy parity, and the pipeline timestamp check still hold.

**[ ]** `npm test` — floor, contract, state, reconciler, gates, copy, contrast, and the new section render all pass.

**[ ]** `npm run lint` — eslint clean.

**[ ]** `npm run build` — Next.js 16 production build succeeds.

**[ ]** `git diff --stat -- lib/assessment.ts lib/types.ts tests/assessment.test.ts` — the floor and its locked tests were not edited.

**[ ]** `git status --short -- lib/clinical components/ContractSections.tsx` — the only product change is the view helper, the section component, its test, and the two surface mounts.

 **The plan is not complete until every box is checked and every command passes. If for some reason a step is not possible to complete, mark it with [f] and move on if possible.**

## Notes

### What "architecture so far" means

Read this as three layers that already exist, plus one layer that is specified and not built.

| LAYER                                                                                      | STATUS                                                          | WHERE                                                                          |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------|
| Three surfaces, EN/SW, referral machine, access gate, voice degrade, location-blind Places | On <code>main</code> through PR #7, commit <code>2a42af2</code> | <code>components/</code> , <code>lib/store.tsx</code> , <code>app/api/*</code> |
| Ten-section contract                                                                       | Committed on the branch, <code>2a1812d</code>                   | <code>lib/clinical/contract.ts</code>                                          |
| State, reconciler, adversarial fixtures, release gates, pipeline                           | Written, uncommitted, gates were green on 2026-10-06            | <code>lib/clinical/</code> , <code>evals/</code>                               |
| Screen uses the contract                                                                   | Not built. This plan.                                           | <code>HouseholdSurface</code> , <code>ChpSurface</code>                        |

### Disposition table already frozen

`DISPOSITION_TABLE` in `lib/clinical/contract.ts` is the only map. Do not add a fifth headline.

| DISPOSITION          | ACTION   | HEADLINE KEY      |
|----------------------|----------|-------------------|
| go_now               | ESCALATE | goNow             |
| get_care_today       | ANSWER   | getCareToday      |
| monitor_at_home      | ANSWER   | monitorAtHome     |
| need_one_more_answer | ASK      | needOneMore       |
| escalate             | ESCALATE | goNow, reuse only |
| abstain              | ABSTAIN  | none              |

Private urgency words are a different scale: `self-care`, `see-clinic`, `urgent`. The reconciler ranks them so they cannot downgrade the floor. Ranking is an engineering order, not a clinical severity score. Do not treat the words as synonyms in copy.

### Reconcile rule, already implemented

```

floor = decide(answers).kind
if no candidate: return floor
if candidate rank is missing: return floor, reason candidate_unrecognized
if candidate rank < floor rank: return floor, reason candidate_below_floor
otherwise: return floor, reason candidate_not_approved
fail_open is always false

```

A safer-looking candidate is not applied in this wave. There is no approved clinical rule pack above the demo floor. `UNAPPROVED_CLINICAL_RULES` is `"disabled"`.

### Pipeline shape, already implemented

```

projectEncounter(encounter)
append extra events, if any
floor = decide(encounter.answers)
reconcileWithTelemetry({ floor, candidate: null })
validateClinicalAssessment(ten sections)
renderAssessment(locale) // labels via t(), collapsed: false

```

`rules_version` on this path is `demo-floor/unvalidated`. Keep that string. Do not rename it to something that sounds reviewed.

## Closed until evidence exists

- **CLOSED** Retrieval ingestion. No approved source with publisher, jurisdiction, version, and effective date.
- **CLOSED** Private-lane adapter. Ticket written. Route still cast-only, logs raw model text, no timeout, no floor.
- **CLOSED** SFT, QLoRA, distillation, quantization. No label producer, no data-rights basis, no measured failure that training would change. A case row is not a training example.
- **CLOSED** IMCI/ETAT numeric thresholds. Encoding them without a named clinician and a versioned rule pack would fake approval.
- **CLOSED** JEV. External hosted decision model. Not a Tunza primitive. Not in this tree.
- **CLOSED** Cross-encounter patient state. Would expand what localStorage holds before privacy projections exist.

Opening criterion for each door is in `vault/wave2-ledger.md`. This plan cannot supply that evidence.

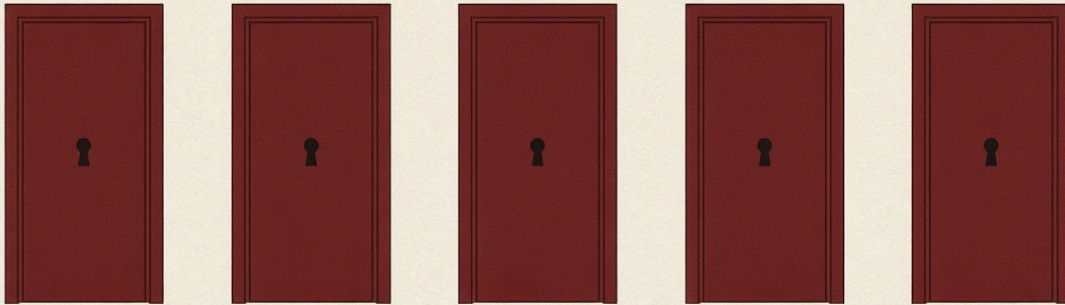
## Stale outside the repo

The second-brain synthesis page still says Tunza is pre-build and that JEV, TypeSafe, and an H100 campaign are not in `~/code/Tunza`. The second sentence is still true: those names are not this architecture. The first sentence is not. The public app and the wave-1 contract are the architecture. A Jacob brief that described JEV and H100 as internal Tunza architecture was already filed as an overclaim. Do not revive it while wiring this slice.

Team context, not an implementation input: Ben Birnbaum owns model and eval architecture; Jacob Hirschfeld owns clinical outreach. Neither role is a substitute for the unsigned clinical review this plan refuses to invent.

## Rejected approaches

- Replacing `decide()` with the pipeline inside the reducer. That would persist a richer clinical object and make the floor harder to lock with the existing assessment tests.
- Rendering one generated explanation. The contract exists so sections cannot collapse.
- Connecting the private Anthropic route in the same change. Its gaps are recorded. Fixing them is a different ticket, and it still would not be a release.
- Calling the result a model. There is nothing to fine-tune.



Closed until evidence.

Training, retrieval, guidelines, the private route, and cross-encounter state stay closed.

## Amendments

None. Append-only after the first execution. Do not rewrite history here.